

Ex. 8: Producer Consumer Using Semaphores

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Karolin@localhost:~  
[Karolin@localhost ~]$ vi producerconsumer.c  
[Karolin@localhost ~]$ gcc producerconsumer.c  
[Karolin@localhost ~]$ cat producerconsumer.c  
#include<stdio.h>  
#include<stdlib.h>  
int mutex=1,full=0,empty=5,x=0;  
int wait(int s){  
    return s--;  
}  
int signal(int s){  
    return ++s;  
}  
void producer(){  
    mutex=wait(mutex);  
    empty=wait(empty);  
    full=signal(full);  
    x++;  
    printf("\nProducer produces item %d",x);  
    mutex=signal(mutex);  
}  
void consumer(){  
    mutex=wait(mutex);  
    full=wait(full);  
    empty=signal(empty);
```

```
    mutex=wait(mutex);  
    full=wait(full);  
    empty=signal(empty);  
    printf("\nConsumer consumes item %d",x);  
    x--;  
    mutex=signal(mutex);  
}  
int main(){  
    int n;  
    printf("\n----Producer Consumer Problem----");  
    printf("\n1.Producer");  
    printf("\n2.Consumer");  
    printf("\n3.Exit");  
    while(1){  
        printf("\nEnter your choice:");  
        scanf("%d",&n);  
        switch(n){  
            case 1:  
                if(mutex==1&&empty!=0){  
                    producer();  
                }  
                else{  
                    printf("\nBuffer is full!! Producer cannot produce.");  
                }  
            case 2:  
                consumer();  
            case 3:  
                return 0;  
        }  
    }  
}
```

```
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}   
break;   
case 2:   
    if(mutex==1&&full!=0){   
        consumer();   
    }   
    else{   
        printf("\nBuffer is empty!! Consumer can   
not consume.");   
    }   
    break;   
case 3:   
    printf("\nExiting program...");   
    printf("\nFinal buffer items produced: %d\n",x);   
    exit(0);   
default:   
    printf("\nInvalid choice. Try again.");   
}   
}   
return 0;   
}   
[Karolin@localhost ~]$ gcc producerconsumer.c -o producerconsumer
```

```
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}   
return 0;   
}   
[Karolin@localhost ~]$ gcc producerconsumer.c -o producerconsumer   
[Karolin@localhost ~]$ ./producerconsumer   
----Producer Consumer Problem----   
1.Producer   
2.Consumer   
3.Exit   
Enter your choice:1   
  
Producer produces item 1   
Enter your choice:1   
  
Buffer is full!! Producer cannot produce.   
Enter your choice:2   
  
Buffer is empty!! Consumer cannot consume.   
Enter your choice:3   
  
Exiting program...   
Final buffer items produced: 1   
[Karolin@localhost ~]$
```